



What's proven, what we checked, what isn't



PROVEN • PUBLISHED

- Physics: rain weakens microwave signals (ITU-R P.838)
- Method: rain from a phone network (Science 2006)
- Scale: country-wide rain maps (PNAS 2013)
- Open-source tools exist (pycomlink, KIT Germany)



OUR CHECK • PUBLIC DATA

- 500 real links, Germany, 11 days, radar as reference
- 0.95 network-wide match, unseen days
- 0.88 typical single link
- 68% of trigger events caught; 18% of our triggers unconfirmed
- Rain / no-rain days: 91% detected, 5% false



NOT YET PROVEN • THE PILOT

- Jio's usable coverage → Gate 1
- Accuracy in Indian monsoon → Gate 2
- Business value inside Reliance → Gate 3
- Who pays → payer test in the pilot
- Insurance-grade triggers → accuracy + actuarial gates, shadow payouts

We know what is proven. We know what isn't. The pilot is built around the unknowns.



Every limit we know of, and our answer

Limitation	Our answer
Jio may not have enough usable microwave links	Coverage audit is Gate 1; radar, gauges and satellite still work without links
Fibre doesn't sense rain	We only use microwave links
A link measures path-averaged rain, not a point	Fusion with radar and gauges turns it into grid estimates
Links alone caught only 68% of trigger events	We never use links alone; accuracy picks the mix
18% of our triggers weren't confirmed by radar	Multi-source confirmation and local calibration
Rain can be confused with equipment faults	Network telemetry, fault flags and cross-link checks
Performance in India is unknown	Back-test Jio's archived Mumbai monsoon data (Gate 2)
1 km resolution isn't guaranteed	Resolution depends on link density and fusion; reported per area
Platforms' willingness to pay is unknown	Prove value inside Reliance first (Gate 3), then sell
Insurance basis risk	Small zones, multi-source trigger, review process
₹300 payout is arbitrary today	Prototype assumption; set by actuarial calibration
₹208 isn't priced on the 10 mm in 3 h trigger	Priced on 25 years of daily IMD risk; the pilot recalibrates the trigger
Towers don't sense heat	IMD stations and satellite feed the same engine
Insurance may never launch	The intelligence business stands on its own